

If a graph has large minimum degree, i.e. everywhere, locally, many edges per vertex, it also has many edges per vertex globally: $\varepsilon(G) = \frac{1}{2}d(G) \geq \frac{1}{2}\delta(G)$. Conversely, of course, its average degree may be large even when its minimum degree is small. However, the vertices of large degree cannot be scattered completely among vertices of small degree: as the next proposition shows, every graph G has a subgraph whose average degree is no less than the average degree of G , and whose minimum degree is more than half its average degree:

[1.4.3]
[7.2.2]

Proposition 1.2.2. *Every graph G with at least one edge has a subgraph H with $\delta(H) > \varepsilon(H) \geq \varepsilon(G)$.*

Proof. To construct H from G , let us try to delete vertices of small degree one by one, until only vertices of large degree remain. Up to which degree $d(v)$ can we afford to delete a vertex v , without lowering ε ? Clearly, up to $d(v) = \varepsilon$: then the number of vertices decreases by 1 and the number of edges by at most ε , so the overall ratio ε of edges to vertices will not decrease.

Formally, we construct a sequence $G = G_0 \supseteq G_1 \supseteq \dots$ of induced subgraphs of G as follows. If G_i has a vertex v_i of degree $d(v_i) \leq \varepsilon(G_i)$, we let $G_{i+1} := G_i - v_i$; if not, we terminate our sequence and set $H := G_i$. By the choices of v_i we have $\varepsilon(G_{i+1}) \geq \varepsilon(G_i)$ for all i , and hence $\varepsilon(H) \geq \varepsilon(G)$.

What else can we say about the graph H ? Since $\varepsilon(K^1) = 0 < \varepsilon(G)$, none of the graphs in our sequence is trivial, so in particular $H \neq \emptyset$. The fact that H has no vertex suitable for deletion thus implies $\delta(H) > \varepsilon(H)$, as claimed. $\square$

1.3 Paths and cycles

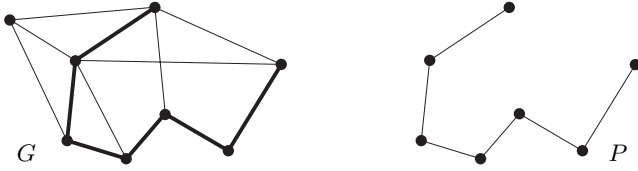
path A *path* is a non-empty graph $P = (V, E)$ of the form

$$V = \{x_0, x_1, \dots, x_k\} \quad E = \{x_0x_1, x_1x_2, \dots, x_{k-1}x_k\},$$

length P^k where the x_i are all distinct. The vertices x_0 and x_k are *linked* by P and are called its *endvertices* or *ends*; the vertices $x_1, \dots, x_{k-1}$ are the *inner* vertices of P . The number of edges of a path is its *length*, and the path of length k is denoted by P^k . Note that k is allowed to be zero; thus, $P^0 = K^1$.

We often refer to a path by the natural sequence of its vertices,³

³ More precisely, by one of the two natural sequences: $x_0 \dots x_k$ and $x_k \dots x_0$ denote the same path. Still, it often helps to fix one of these two orderings of $V(P)$ notationally: we may then speak of things like the ‘first’ vertex on P with a certain property, etc.

Fig. 1.3.1. A path $P = P^6$ in G

writing, say, $P = x_0x_1 \dots x_k$ and calling P a path from x_0 to x_k (as well as *between* x_0 and x_k).

For $0 \leq i \leq j \leq k$ we write

$$Px_i := x_0 \dots x_i$$

$$x_iP := x_i \dots x_k$$

$$x_iPx_j := x_i \dots x_j$$

and

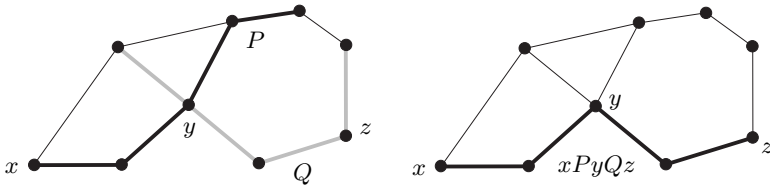
$$\overset{\circ}{P} := x_1 \dots x_{k-1}$$

$$P\overset{\circ}{x}_i := x_0 \dots x_{i-1}$$

$$\overset{\circ}{x}_iP := x_{i+1} \dots x_k$$

$$\overset{\circ}{x}_iP\overset{\circ}{x}_j := x_{i+1} \dots x_{j-1}$$

for the appropriate subpaths of P . We use similar intuitive notation for the concatenation of paths; for example, if the union $Px \cup xQy \cup yR$ of three paths is again a path, we may simply denote it by $PxQyR$.

Fig. 1.3.2. Paths P , Q and $xPyQz$

Given sets A, B of vertices, we call $P = x_0 \dots x_k$ an A - B path if $V(P) \cap A = \{x_0\}$ and $V(P) \cap B = \{x_k\}$. As before, we write a - B path rather than $\{a\}$ - B path, etc. Two or more paths are *independent* if none of them contains an inner vertex of another. Two a - b paths, for instance, are independent if and only if a and b are their only common vertices.

A non-trivial path P is an A -path for a set A of vertices if P has its ends but no inner vertex in A . It is an H -path for a graph H if it is a $V(H)$ -path and, if it has length 1, its edge does not lie in H .

$xPy, \overset{\circ}{P}$

$PxQyR$

A - B path

independent

A -path

H -path

cycle

length

 C^k girth $g(G)$ circum-
ference

chord

induced
cycle

If $P = x_0 \dots x_{k-1}$ is a path and $k \geq 3$, then the graph $C := P + x_{k-1}x_0$ is called a *cycle*. As with paths, we often denote a cycle by its (cyclic) sequence of vertices; the above cycle C might be written as $x_0 \dots x_{k-1}x_0$. The *length* of a cycle is its number of edges (or vertices); the cycle of length k is called a *k-cycle* and denoted by C^k .

The minimum length of a cycle (contained) in a graph G is the *girth* $g(G)$ of G ; the maximum length of a cycle in G is its *circumference*. (If G does not contain a cycle, we set the former to ∞ , the latter to zero.)

An edge which joins two vertices of a cycle but is not itself an edge of the cycle is a *chord* of that cycle. Thus, an *induced cycle* in G , a cycle in G forming an induced subgraph, is one that has no chords (Fig. 1.3.3).

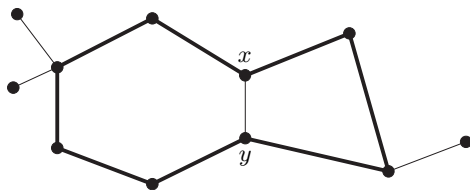


Fig. 1.3.3. A cycle C^8 with chord xy , and induced cycles C^6, C^4

If a graph has large minimum degree, it contains long paths and cycles (see also Exercise 9):

[1.4.3]
[7.2.2]

Proposition 1.3.1. Every graph G contains a path of length $\delta(G)$ and a cycle of length at least $\delta(G) + 1$ (provided that $\delta(G) \geq 2$).

Proof. Let $x_0 \dots x_k$ be a longest path in G . Then all the neighbours of x_k lie on this path (Fig. 1.3.4). Hence $k \geq d(x_k) \geq \delta(G)$. If $i < k$ is minimal with $x_i x_k \in E(G)$, then $x_i \dots x_k x_i$ is a cycle of length at least $\delta(G) + 1$. $\square$

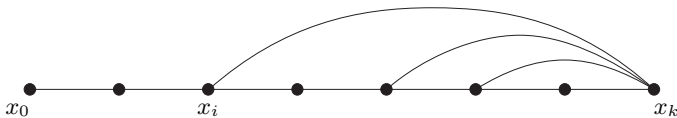


Fig. 1.3.4. A longest path $x_0 \dots x_k$, and the neighbours of x_k

Minimum degree and girth, on the other hand, are not related (unless we fix the number of vertices): as we shall see in Chapter 11, there are graphs combining arbitrarily large minimum degree with arbitrarily large girth.

distance
 $d(x, y)$ diameter
 $\text{diam}(G)$

The *distance* $d_G(x, y)$ in G of two vertices x, y is the length of a shortest x - y path in G ; if no such path exists, we set $d(x, y) := \infty$. The greatest distance between any two vertices in G is the *diameter* of G , denoted by $\text{diam}(G)$. Diameter and girth are, of course, related:

Proposition 1.3.2. *Every graph G containing a cycle satisfies $g(G) \leq 2 \operatorname{diam}(G) + 1$.*

Proof. Let C be a shortest cycle in G . If $g(G) \geq 2 \operatorname{diam}(G) + 2$, then C has two vertices whose distance in C is at least $\operatorname{diam}(G) + 1$. In G , these vertices have a lesser distance; any shortest path P between them is therefore not a subgraph of C . Thus, P contains a C -path xPy . Together with the shorter of the two x - y paths in C , this path xPy forms a shorter cycle than C , a contradiction. $\square$

A vertex is *central* in G if its greatest distance from any other vertex is as small as possible. This distance is the *radius* of G , denoted by $\operatorname{rad}(G)$. Thus, formally, $\operatorname{rad}(G) = \min_{x \in V(G)} \max_{y \in V(G)} d_G(x, y)$. As one easily checks (exercise), we have

central

radius
 $\operatorname{rad}(G)$

$$\operatorname{rad}(G) \leq \operatorname{diam}(G) \leq 2 \operatorname{rad}(G).$$

Diameter and radius are not related to minimum, average or maximum degree if we say nothing about the order of the graph. However, graphs of large diameter and minimum degree must be large (larger than forced by each of the two parameters alone; see Exercise 10), and graphs of small diameter and maximum degree must be small:

Proposition 1.3.3. *A graph G of radius at most k and maximum degree at most $d \geq 3$ has fewer than $\frac{d}{d-2}(d-1)^k$ vertices.*

[9.4.1]
[9.4.2]

Proof. Let z be a central vertex in G , and let D_i denote the set of vertices of G at distance i from z . Then $V(G) = \bigcup_{i=0}^k D_i$. Clearly $|D_0| = 1$ and $|D_1| \leq d$. For $i \geq 1$ we have $|D_{i+1}| \leq (d-1)|D_i|$, because every vertex in D_{i+1} is a neighbour of a vertex in D_i (why?), and each vertex in D_i has at most $d-1$ neighbours in D_{i+1} (since it has another neighbour in D_{i-1}). Thus $|D_{i+1}| \leq d(d-1)^i$ for all $i < k$ by induction, giving

$$|G| \leq 1 + d \sum_{i=0}^{k-1} (d-1)^i = 1 + \frac{d}{d-2} ((d-1)^k - 1) < \frac{d}{d-2} (d-1)^k.$$

 $\square$

Similarly, we can bound the order of G from below by assuming that both its minimum degree and girth are large. For $d \in \mathbb{R}$ and $g \in \mathbb{N}$ let

$$n_0(d, g) := \begin{cases} 1 + d \sum_{i=0}^{r-1} (d-1)^i & \text{if } g =: 2r+1 \text{ is odd;} \\ 2 \sum_{i=0}^{r-1} (d-1)^i & \text{if } g =: 2r \text{ is even.} \end{cases}$$

It is not difficult to prove that a graph of minimum degree δ and girth g has at least $n_0(\delta, g)$ vertices (Exercise 7). Interestingly, one can obtain the same bound for its average degree:

Theorem 1.3.4. (Alon, Hoory & Linial 2002)

Let G be a graph. If $d(G) \geq d \geq 2$ and $g(G) \geq g \in \mathbb{N}$ then $|G| \geq n_0(d, g)$.

One aspect of Theorem 1.3.4 is that it guarantees the existence of a short cycle compared with $|G|$. Using just the easy minimum degree version of Exercise 7, we get the following rather general bound:

[2.3.1] **Corollary 1.3.5.** If $\delta(G) \geq 3$ then $g(G) < 2 \log |G|$.

Proof. If $g := g(G)$ is even then

$$n_0(3, g) = 2 \frac{2^{g/2} - 1}{2 - 1} = 2^{g/2} + (2^{g/2} - 2) > 2^{g/2},$$

while if g is odd then

$$n_0(3, g) = 1 + 3 \frac{2^{(g-1)/2} - 1}{2 - 1} = \frac{3}{\sqrt{2}} 2^{g/2} - 2 > 2^{g/2}.$$

As $|G| \geq n_0(3, g)$, the result follows. $\square$

walk

A *walk* (of length k) in a graph G is a non-empty alternating sequence $v_0 e_0 v_1 e_1 \dots e_{k-1} v_k$ of vertices and edges in G such that $e_i = \{v_i, v_{i+1}\}$ for all $i < k$. If $v_0 = v_k$, the walk is *closed*. If the vertices in a walk are all distinct, it defines an obvious path in G . In general, every walk between two vertices contains⁴ a path between these vertices (proof?).

1.4 Connectivity

connected

A graph G is called *connected* if it is non-empty and any two of its vertices are linked by a path in G . If $U \subseteq V(G)$ and $G[U]$ is connected, we also call U itself connected (in G). Instead of ‘not connected’ we usually say ‘disconnected’.

[1.5.2] **Proposition 1.4.1.** The vertices of a connected graph G can always be enumerated, say as $v_1, \dots, v_n$, so that $G_i := G[v_1, \dots, v_i]$ is connected for every i .

⁴ We shall often use terms defined for graphs also for walks, as long as their meaning is obvious.